

ASSIGNMENT #02

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, February 20th TA's Mailbox

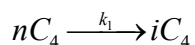
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Plant Wide Process Control.

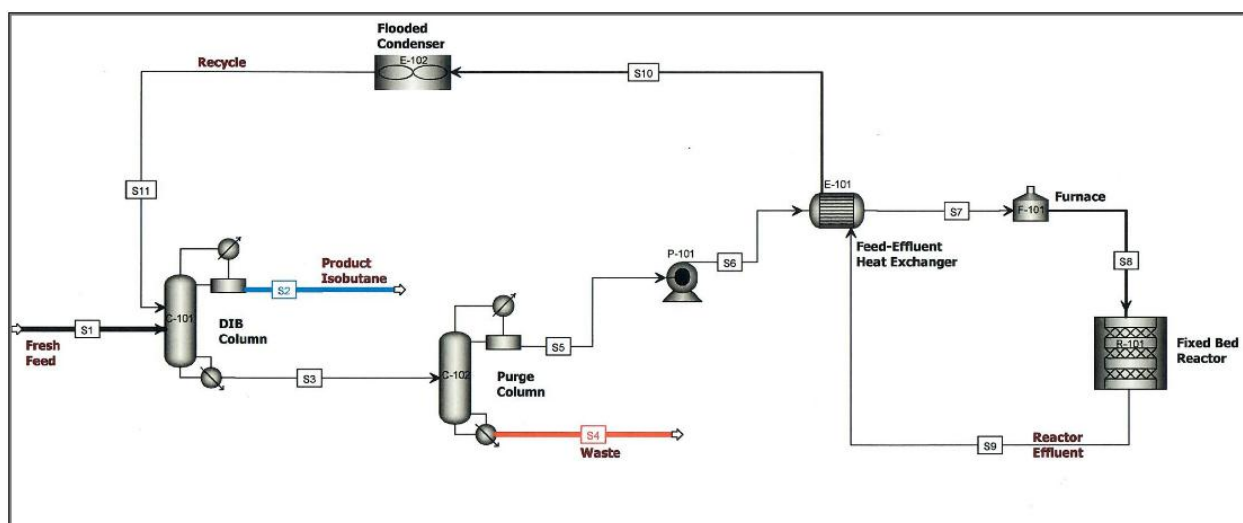
A process flowsheet diagram for the production of iso-butane via the isomerization of n-butane is shown in the Figure below: The isomerization reaction takes place in the fixed bed reactor with pressure drop equal to 15 psi. The kinetics of the reaction are shown below:



$$R_1 = k_{10} \exp\left\{-\frac{E}{RT}\right\} C_{nC_4}$$

$$E = 30,000 \text{ Btu} / \text{lbmol}$$

$$k_{10} = 4 \times 10^8 \text{ l} / \text{hr}$$



To better understand the process, open the attached ASPEN simulation and carefully study the operating conditions of all unit operations. The distillate product from the DeIsoButanizer column (DIB) has a specification of 2 mol% n-butane and is directed to a storage tank. The plant uses cooling water from near-by cooling tower designed with 75% efficiency with hot water inlet temperature of 120 °F and air ambient temperature of 90 °F with relative humidity 70%. A steam plant from a nearby facility supplies saturated steam (with quality 0.95) at three different levels of pressure: low pressure at 50 psig, medium pressure at 150 psig and high pressure at 600 psig. Natural gas from an existing pipeline is used for the fired heater that heats the feed to the reactor. The heat transfer coefficients in all the heat exchangers can be taken as the middle values of the ranges that are listed in the **CHE Calculator**.

Assume that the pressure drop in the heat exchanger is 10 psi both on the tube side and 5 psi on the shell side and the pressure drop in the fired heater is 60 psi. The pressure drop in the liquid line from the pump discharge to the inlet of the fired heater is approximately 8 psi.

This assignment has three parts:

Part A: Flowsheet Modifications

1. First you must modify the simulation to obtain the product specifications and the diameters of the distillation columns. That will help you calculate the height of the liquid in the column sump. Some modifications will also be need to accommodate the additional information on pressure drop given in the problem description above.
2. Second you must create three utilities (cooling water, steam and natural gas) and attach them to the appropriate unit operations. Utility consumption will be used to calculate the steady-state values of the manipulated variables for the process control loops that you will identify in **Part B**.

Part B: Identification and Design of Process Control Loops

Following the procedure that I have outlined in the lectures, calculate the Control Degrees of Freedom for this process and design all process control loops using the additional information outlined below.

Data and Assumptions

1. Both refluxes in the **DIB** column and the **Purge** column are on flow control
2. Residence time in the reflux accumulation drums: 4 **min**
3. Residence time in the distillation column sumps: 2 **min**
4. The waste stream is also on flow control due to environmental restrictions
5. Neither the fresh feed nor the product flow rates are fixed by other plant considerations
6. Heating value of natural gas used in the fired heater: 50.0 **MJ/kg**
7. The heat capacity of water can be taken equal to 4.18 **kJ/(kg C)**
8. Temperature rise in the cooling water is limited to 11 °C

Process Variables Units for Reporting

1. Liquid flow rates must be given in **gpm**
2. Gas flow rates in **lb/hr**
3. Temperature in **deg F**
4. Pressure in **psia**
5. Height in **ft**

For each control loop you have identified write a summary similar to the template shown below:

Template: Design of Process Control Loops

Loop #	Controller Description	Controlled Variable: CV	Set Point Value: SP	Manipulated Variable: MV	Steady-State Normal Value:
1	DIB Column Reflux Flow	Reflux Flow	7.78 Reflux Ratio	Reflux Flow	7.78 Reflux Ratio

2	Fresh Feed Flow	Feed Flow	33955.6 lb/hr	Feed Flow	33955.6 lb/hr
3	DIB Column Condenser	Cooling Water Flow	2.03×10^6 lb/hr CW	Cooling Water Flow	2.03×10^6 lb/hr CW
4	Purge Column Condenser	Cooling Water Flow	737313 lb/hr CW	Cooling Water Flow	737313 lb/hr CW
5	DIB Column Reboiler	Heat Flow	38205.5 lb/hr LP Steam	Heat Flow	38205.5 lb/hr LP Steam
6	Purge Column Reboiler	Heat Flow	12386.5 LP Steam	Heat Flow	12386.5 LP Steam
7	Purge Column Reflux Flow	Reflux Flow	0.8 Reflux Ratio	Reflux Flow	0.8 Reflux Ratio
8	DIB Column Liquid Height	DIB Column Liquid Height	52570.2 lb/hr	Purge Column Feed Flow	52570.2 lb/hr
9	Fixed Bed Reactor Temperature	Heat Flow in Furnace Pump before needed	50.0 MJ/kg	Heat Flow in Furnace	50.0 MJ/kg
10	Waste	Waste flow	2178.6 lb/hr	Waste Flow	2178.6 lb/hr
11	Product Isobutane	Isobutane Flow	31777 lb/hr	Isobutane Flow	31777 lb/hr
12	Flow Rate S6 to S11	Pump Energy	66.77 kW	Pump Energy	66.77 kW
13	Flooded Condenser	Cooling Water Flow	423306 lb/hr	Cooling Water Flow	423306 lb/hr
14	N/A				
15	N/A				

Reading through Dr. C's second chapter of his textbook, the Set Point value and the simulation steady state value are equivalent. This process sees the same issue as report one, where it is difficult to control the steam loop that shares a heat exchanger. There should be some other form of equipment to potentially change the flow rate in streams 7, 8 or 9 to not lose control of the entire process. When "hold up" values within the distillation columns can no longer be used as an option to "hold" material, the utility streams become the only flow rate that can be altered significantly. The only true thing that can be altered in any process control scenario is the flow rate.

Part C: Process Simulation Calculations

From pilot plant studies it was recommended to increase the reactor feed temperature by 10 °F in order to increase production rate. Change the set point of the temperature controller and answer the following questions after carrying out the simulation with the new reactor feed temperature.

1. Briefly explain your logic in setting up the simulation for the change in the temperature loop set point value from 390 deg F to 400 deg F. Did you have to change the set points of any other controllers? If yes provide a brief explanation as to why and then enter their new values in the template shown below.
2. Run the simulation in order to calculate the new steady-state values of the manipulated values for all controllers and enter the new values in the template shown below

Template: Simulation Calculations after Set Point Change

Loop #	Controller Description	Controlled Variable: CV	Old Set Point: SP	New Set Point: SP	MV	Old Steady State Value	New Steady State Value
1	DIB Column Reflux Flow	Reflux Flow	7.78 Reflux Ratio	7.79 Reflux Ratio	Reflux Flow	7.78 Reflux Ratio	7.79 Reflux Ratio
2	Fresh Feed Flow	Feed Flow	33955.6 lb/hr	33955.6 lb/hr	Feed Flow	33955.6 lb/hr	33955.6 lb/hr
3	DIB Column Condenser	Cooling Water Flow	2.03*10 ⁶ lb/hr CW	2.03*10 ⁶ lb/hr CW	Cooling Water Flow	2.03*10 ⁶ lb/hr CW	2.03*10 ⁶ lb/hr CW
4	Purge Column Condenser	Cooling Water Flow	737313 lb/hr CW	737313 lb/hr CW	Cooling Water Flow	737313 lb/hr CW	737313 lb/hr CW
5	DIB Column Reboiler	Heat Flow	38205.5 lb/hr LP Steam	38205.5 lb/hr LP Steam	Heat Flow	38205.5 lb/hr LP Steam	38205.5 lb/hr LP Steam
6	Purge Column Reboiler	Heat Flow	12386.5 lb/hr LP Steam	12386.5 lb/hr LP Steam	Heat Flow	12386.5 lb/hr LP Steam	12386.5 lb/hr LP Steam
7	Purge Column Reflux Flow	Reflux Flow	0.8 Reflux Ratio	0.8 Reflux Ratio	Reflux Flow	0.8 Reflux Ratio	0.8 Reflux Ratio

8	Purge Column Feed Flow	DIB Column Liquid Height	52570.2 lb/hr	52570.4 lb/hr	Purge Column Feed Flow	52570.2 lb/hr	52570.4 lb/hr
9	Furnace Temperature	Heat Flow in Furnace Pump before needed	50.0 MJ/kg	Unable to change due to nature of heater	Heat Flow in Furnace	50.0 MJ/kg	Unable to change due to nature of heater
10	Waste	Waste flow	2178.6 lb/hr	2178.6 lb/hr	Waste Flow	2178.6 lb/hr	2178.6 lb/hr
11	Product Isobutane	Isobutane Flow	31777 lb/hr	31777 lb/hr	Isobutane Flow	31777 lb/hr	31777 lb/hr
12	Flow to S6	Pump Energy	66.77 kW	66.77 kW	Pump Energy	66.77 kW	66.77 kW
13	Flooded Condenser	Cooling Water Flow	423306 lb/hr	423298 lb/hr	Cooling Water Flow	423306 lb/hr	423298 lb/hr
14	N/A						
15	N/A						

Items with bold values are those that changed from the initial set point. Note that the product stream did not increase in flow within the simulation. While I changed the S8 temperature within Aspen, the more accurate thing to do might have been to increase the amount of utility flow of the furnace.

- I did go to Dr. C's office hours for clarification on the utility specs, what I have turned in is what he told me to do.